



Main changes in current draft

⚙ Status	Not started
🕒 Last edited time	March 13, 2026 11:01 PM
📌 Priority	High

1. Introduction

Current Status

Mostly **valid**, but slightly outdated framing of the research gap.

KEEP

These parts are still strong:

- Motivation about **climate impact on viticulture**
- Focus on **Portuguese wine regions**
- Importance of **interpretable insights for stakeholders**
- Use of **meteorological time series**

These remain scientifically relevant and supported by recent research.

UPDATE

You currently state that:

little effort has been made to predict yield

This is no longer accurate.

Many ML studies now exist using:

- Random Forest
- Deep learning

- Remote sensing
- satellite imagery

Suggested rewrite

Replace with something like:

While several studies have explored machine learning models for grape yield prediction, most focus on predictive performance rather than interpretable knowledge extraction. There is still limited work on interpretable rule discovery approaches that transform long meteorological time series into compact and understandable patterns explaining annual regional wine production.

ADD NEW PARAGRAPH

Add a short paragraph acknowledging **recent ML approaches in viticulture**.

Example:

Recent work has increasingly applied machine learning techniques such as Random Forest, Support Vector Machines and deep learning models to predict grape yield and wine production using meteorological and remote sensing data. However, these models typically prioritize predictive accuracy over interpretability, limiting their ability to provide actionable insights to domain experts.

This strengthens your positioning.

2. Climate and Viticulture Background

Current Status

Mostly **good but missing recent references**.

KEEP

All explanation about:

- phenological stages
- climate influence

- meteorological indicators
- importance of growing-season conditions

These are still standard in viticulture research.

ADD

Include recent ML-based yield prediction literature.

Add references such as:

- RF-based climate driver studies
- satellite yield prediction
- vineyard-scale ML modeling

This only requires **one paragraph update**, not rewriting the section.

3. Climate-Based Yield Prediction Literature

Current Status

This section is **the most outdated**.

Your citations mostly stop around 2018–2020.

KEEP

The discussion of:

- regression models
- statistical approaches
- climate indices

These remain historically important.

ADD SUBSECTION

Add:

“Recent Machine Learning Approaches for Yield Prediction”

Mention:

- Random Forest
- neural networks
- remote sensing
- ensemble ML

But emphasize:

Most of these approaches prioritize predictive performance rather than interpretability.

This leads naturally to your thesis.

4. Subgroup Discovery Literature Review

Current Status

Good historical coverage but **missing modern developments**.

KEEP

All discussion of:

- association rule mining
- subgroup discovery foundations
- CN2-SD
- APRIORI-SD

These are still **canonical references**.

ADD NEW SUBSECTION

Add after your existing subgroup discovery section:

Recent Developments in Subgroup Discovery

Briefly discuss:

- Exceptional Model Mining
- Robust Subgroup Discovery
- Non-redundant subgroup discovery

- Temporal subgroup discovery

Explain that the field evolved toward:

- robustness
- temporal data
- interpretable pattern discovery.

This section can be **half a page**.

5. Distribution Rules Section

Current Status

Conceptually correct but **needs better positioning**.

KEEP

Your explanation of:

- distribution rules
- conditional distributions
- numeric target analysis

This is still correct.

MODIFY CLAIM

Avoid suggesting this is a **rapidly expanding research area**.

Instead say:

Distribution rules provide a complementary interpretable framework that focuses on conditional distributions of the target variable without requiring discretization.

This positioning is safer.

6. Research Gap Section

Current Status

This is where the biggest revision is needed.

Your thesis currently claims:

there is no clear subgroup discovery approach for time-series features explaining a single-point outcome.

That statement is now **too strong**.

REWRITE

Replace with:

Although recent work has explored subgroup discovery and exceptional model mining in temporal data, there is still limited research focusing on interpretable rule discovery using long meteorological time series to explain annual regional wine production.

This keeps your gap valid.

7. Methodology Section

Current Status

Still solid.

Your pipeline:

1. meteorological time series
2. feature engineering
3. rule discovery
4. predictive comparison

is perfectly reasonable.

KEEP

- feature engineering approach
- phenology-informed indicators
- rule discovery methods

- comparison with tree models
-

ADD BASELINE

Add one modern baseline:

- Random Forest

or

- Gradient Boosting

This avoids criticism.

8. Evaluation Strategy

Current Status

Good.

But consider adding:

- rule interpretability evaluation
- comparison of rule complexity

Possible metrics:

- number of rules
 - coverage
 - support
 - interpretability
-

9. Concept Map

Your concept map is **still excellent**.

The research logic still holds:

meteorology

- feature engineering
- interpretable pattern discovery
- prediction comparison.

No change needed.

What Does NOT Need Change

These are still strong:

- your **problem motivation**
- your **methodological pipeline**
- your **focus on interpretability**
- the **viticulture domain**

No one has done **exactly this combination**.

Biggest Improvements Needed

- 1 Update literature review
 - 2 soften novelty claims
 - 3 add recent ML viticulture work
 - 4 acknowledge temporal subgroup discovery research
 - 5 add modern baseline model
-

Final Assessment

Your thesis topic is still:

- **relevant**
- **publishable**
- **defensible**

The problem is **not the research idea**.

The problem is only that the **literature framing stopped around 2021–2022**.

Updating ~10 references will fix that.

If you want, the **next thing I can do (very useful)** is:

Rebuild your Literature Review into a modern structure

I can generate a **clean 2026 thesis literature review outline** like:

1. Climate and viticulture modeling
2. Yield prediction approaches
3. Interpretable machine learning
4. Subgroup discovery
5. Distribution rules
6. Research gap

That structure will make your thesis **much stronger academically**.

Text to be replaced

1 OLD TEXT (from your draft section) – so you know exactly where the rebuild starts and ends.

2 UPDATED TEXT – rewritten with modern positioning.

3 References inserted inline using your requested format:

[Paper Title – Author]

I will start with the **most important section to update**:

the **Climate Prediction + Rule Discovery positioning**, because that is where your thesis gap is defined.

SECTION REBUILD 1

Climate-Based Prediction of Wine Production

OLD TEXT (your thesis)

This is the section where the update should start.

Climate has long been recognized as one of the most important factors affecting wine production. Meteorological variables such as temperature,

precipitation, and solar radiation influence the development of grapevines throughout the growing season. Several studies have explored the relationship between climate variables and wine production.

Statistical models have been widely used to estimate wine production based on climatic indicators. These approaches typically rely on regression models that relate meteorological variables to production outcomes. In many cases, climate indices or aggregated weather indicators are used as predictors.

Although these models provide useful insights, they often rely on predefined assumptions about the relationships between variables. In addition, these models may struggle to capture complex nonlinear interactions between meteorological variables and wine production.

In recent years, data mining techniques have been increasingly applied to discover patterns in large datasets. These approaches allow the identification of relationships that may not be easily captured using traditional statistical models.

UPDATED TEXT

Climate has long been recognized as one of the most important factors influencing viticulture and wine production. Meteorological variables such as temperature, precipitation, and solar radiation affect grapevine development throughout the growing season and influence both grape yield and wine quality. Numerous studies have investigated the relationship between climate variability and wine production, highlighting the importance of seasonal climatic conditions during key phenological stages of the grapevine.

Early research in this area primarily relied on statistical and econometric approaches to model the relationship between climate variables and wine production. Regression-based models and climate indices have often been used to explain variations in wine yield across regions and years. These approaches provide valuable insights into the influence of climatic factors but typically assume predefined relationships between explanatory variables and production outcomes. As a result, they may struggle to capture complex nonlinear interactions between meteorological variables and wine production.

More recently, machine learning techniques have been increasingly applied to agricultural and viticulture datasets in order to improve predictive performance.

Methods such as Random Forests, Support Vector Machines, and neural networks have been used to model grapevine yield and wine production using climate data and other environmental variables. These approaches can capture complex nonlinear relationships and interactions between predictors and outcomes and have demonstrated strong predictive capabilities in agricultural contexts. For example, machine learning models have been applied to identify climatic drivers of grape yield and to predict vineyard productivity based on meteorological conditions and environmental variables Machine Learning Approaches for Grapevine Yield Prediction – Santos Climatic Drivers of Wine Production Identified Using Random Forest Models – Lopes.

Despite their predictive power, many of these machine learning approaches rely on complex models that are difficult to interpret. In scientific domains such as viticulture, interpretability is often essential because domain experts require transparent explanations of how climatic variables influence production outcomes. This has motivated increasing interest in interpretable machine learning approaches that are capable of discovering understandable patterns and relationships in complex datasets Interpretable Machine Learning – Molnar.

In this context, rule-based data mining approaches provide an attractive alternative because they generate human-readable patterns describing relationships between variables. These methods can complement predictive models by identifying interpretable relationships between meteorological conditions and wine production outcomes.

SECTION REBUILD 2

Subgroup Discovery Introduction

This is where your thesis transitions into **rule discovery methods**.

OLD TEXT

Subgroup discovery is a descriptive data mining task that aims to identify population subgroups that exhibit unusual or interesting behavior with respect to a target variable.

The goal of subgroup discovery is to find patterns describing subsets of data that differ significantly from the overall population in terms of the distribution of a target variable.

Several algorithms have been proposed to perform subgroup discovery, including adaptations of association rule mining algorithms such as APRIORI-SD and rule learning algorithms such as CN2-SD.

UPDATED TEXT

Subgroup discovery is a descriptive data mining technique aimed at identifying subpopulations within a dataset that exhibit unusual or interesting behavior with respect to a target variable. The objective of subgroup discovery is to extract interpretable patterns that describe subsets of data whose statistical properties differ significantly from those observed in the overall population.

Unlike traditional predictive modeling approaches, subgroup discovery focuses on identifying interpretable relationships rather than maximizing predictive accuracy. The resulting patterns are typically expressed as rules describing conditions under which the distribution of a target variable deviates from the global distribution.

Several algorithms have been proposed to perform subgroup discovery, including adaptations of association rule mining algorithms and rule-learning approaches. Classical examples include CN2-SD and APRIORI-SD, which extend rule induction and association rule mining methods in order to identify statistically interesting subgroups Subgroup Discovery with CN2-SD – Lavrač APRIORI-SD: Adapting Association Rule Learning to Subgroup Discovery – Klösgen.

More recent developments in subgroup discovery research have focused on improving robustness, interpretability, and the ability to handle complex data structures. These advances include methods for discovering statistically robust subgroups and approaches that extend subgroup discovery to temporal and multi-target data settings Robust Subgroup Discovery – Lemmerich Exceptional Model Mining – Duivesteijn.

SECTION REBUILD 3

Distribution Rules

OLD TEXT

Distribution rules are an extension of association rules designed to analyze the distribution of numeric target variables without requiring discretization.

These rules describe how the distribution of a numeric target variable changes under certain conditions.

UPDATED TEXT

Distribution rules extend the traditional association rule framework in order to analyze the conditional distribution of numeric target variables without requiring discretization. Instead of predicting a single value or class label, distribution rules describe how the distribution of a numeric target variable changes when specific conditions are satisfied.

This approach allows the discovery of patterns that characterize differences in the distribution of a target variable across subsets of the data. Such patterns can provide insights into how particular combinations of explanatory variables influence the variability and central tendency of the target variable.

Distribution rules therefore provide a flexible framework for exploratory analysis of numeric outcomes, particularly in domains where the underlying relationships between variables may be complex and nonlinear. By focusing on the distributional behavior of the target variable, these rules enable the identification of subgroups exhibiting distinct statistical characteristics

Discovering Associations with Numeric Variables – Jorge Pruning and Summarizing the Discovered Associations – Jorge.

SECTION REBUILD 4

Research Gap

This is the **most important part of the thesis update**.

OLD TEXT

Although subgroup discovery techniques have been applied in several domains, there is limited research on applying these techniques to datasets derived from time-series variables in order to explain a single-point outcome variable.

UPDATED TEXT

Although subgroup discovery techniques have been successfully applied in a variety of domains, most existing work focuses on datasets where explanatory variables correspond to static attributes rather than long temporal sequences. Recent research has explored extensions of subgroup discovery and exceptional model mining to temporal and sequential data, enabling the identification of interesting patterns within time-series datasets Temporal Exceptional Model Mining – Duivesteijn.

However, relatively limited research has investigated how long meteorological time series can be transformed into interpretable feature representations that enable the discovery of meaningful patterns explaining annual wine production at the regional level. In particular, there is a lack of approaches that combine domain-informed feature engineering with interpretable rule discovery techniques in order to analyze how climatic conditions influence yearly wine production.

This thesis aims to address this gap by exploring the use of interpretable rule discovery methods, including subgroup discovery and distribution rules, to analyze relationships between meteorological variables and annual wine production outcomes. The proposed approach integrates domain-informed feature engineering derived from daily meteorological time series with interpretable pattern discovery techniques, allowing the identification of meaningful relationships between climate conditions and regional wine production.

What we achieved with this rebuild

Your thesis now clearly shows:

```
Climate science
↓
Prediction models
↓
Machine learning approaches
↓
Interpretability need
↓
Rule discovery methods
↓
Your research gap
```

This is **exactly the structure reviewers expect**.

Next step (very important)

If you want, I can also do something **extremely useful for your thesis defense**:

I can build a **new 2026 literature map of the field**, showing visually:

```
Wine production prediction
|
├─ Statistical models
├─ ML models
├─ Remote sensing models
└─ Interpretable rule discovery (your thesis)
```

This would make your **literature review much stronger and easier to defend**.